Tutorial 11 Solution

Problem 1

Consider the following nonlinear programming problem

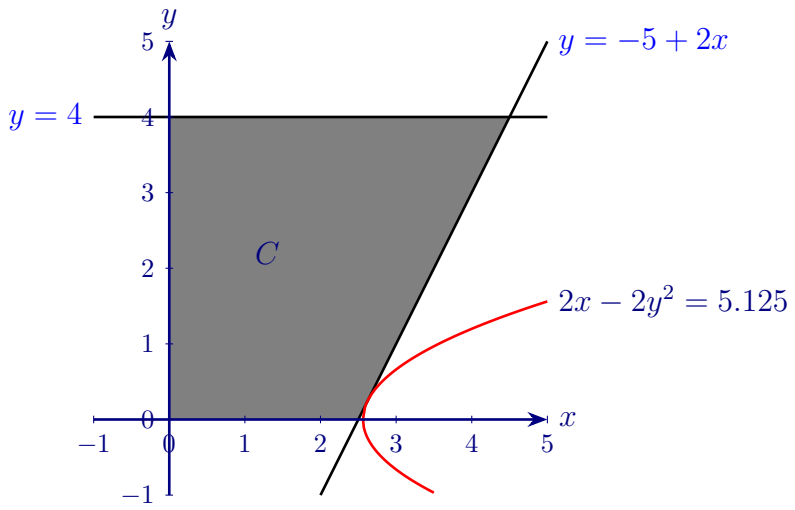
$$\max 2x - 2y^2 \qquad \text{s.t.} \quad 2x - y \leq 5,\ y \leq 4,\ x \geq 0,\ y \geq 0.$$

(a) Try to solve (at least roughly) the problem graphically.

(b) Solve the problem using the KKT approach.

Solution:

(a) The level curve of the objective function of level C is defined by the equation $x = y^2 + \frac{1}{2}C$. You have to draw quite precisely to see the exact solution: it is $(x^*, y^*) = (21/8, 1/4)$.



(b) Because the constraint functions are all affine, the KKT conditions are necessary.

The KKT conditions are

$$\begin{aligned} 2 - 2\lambda_1 \leq 0, \quad & x \geq 0, \quad & x(2 - 2\lambda_1) = 0 \\ -4y + \lambda_1 - \lambda_2 \leq 0, \quad & y \geq 0, \quad & y(-4y + \lambda_1 - \lambda_2) = 0 \\ 2x - y - 5 \leq 0, \quad & \lambda_1 \geq 0, \quad & \lambda_1(2x - y - 5) = 0 \\ y - 4 \leq 0, \quad & \lambda_2 \geq 0, \quad & \lambda_2(y - 4) = 0 \end{aligned}$$

We see that

- $x > 0$ and $\lambda_1 = 1$: Because $2 - 2\lambda_1 \leq 0$, we must have $\lambda_1 > 0$. Hence by complementary slackness $[\lambda_1(2x - y - 5) = 0]$, $2x - y = 5$. But this implies that $x = 0$: if $x = 0$, $2(0) - y = 5$ and thus $y = -5 < 0$, a contradiction. Hence $x > 0$; by complementary slackness $[x(2 - 2\lambda_1) = 0]$, we have $\lambda_1 = 1$.

- Now it remains to discuss the signs of y and λ_2. Suppose $y = 0$, then Thus $y > 0$: Suppose that $y = 0$. Then $-4y + \lambda_1 - \lambda_2 \leq 0$ becomes $1 = \lambda_1 \leq \lambda_2$, so $\lambda_2 > 0$. Again by complementary slackness [$\lambda_2(y-4) = 0$], we have $y = 4$, a contradiction.
- Thus, we must have $y > 0$. By complementary slackness, $-4y + \lambda_1 - \lambda_2 = 0$. Now suppose $\lambda_2 = 0$. Then $y = 1/4$; since $2x - y = 5$, $x = 21/8$. Thus, $(x^*, y^*, \lambda_1^*, \lambda_2^*) = (21/8, 1/4, 1, 0)$ satisfies the KKT conditions.
- Now suppose $\lambda_2 > 0$. By complementary slackness, $y = 4$. Then $-4y + \lambda_1 - \lambda_2 = 0$ yields (since $y = 4$ and $\lambda_1 = 1$) $\lambda_2 = -15 < 0$, a contradiction.

Therefore, the only point satisfying the KKT conditions is $(x^*, y^*, \lambda_1^*, \lambda_2^*) = (21/8, 1/4, 1, 0)$. Since the objective function is continuous and the constraint set is compact, the extreme value theorem applies, and therefore a solution exists. Then since the KKT conditions are necessary; that is, every solution must (together with some $\lambda_1^* \geq 0$ and $\lambda_2^* \geq 0$) satisfy the KKT conditions, we can conclude that $(x^*, y^*) = (21/8, 1/4)$ is the unique solution.

Remark. As illustrated by this question, after you wrote down the KKT conditions, it is usually useful to stare at the conditions and think about whether there are conditions that offer clear guidance of the signs of the variables. Here, the condition $2 - 2\lambda_1 \leq 0$ indicates that $\lambda_1 > 0$.

Problem 2

Let $u : \mathbb{R}^2_+ \to \mathbb{R}$ given by $u(x, y) = \sqrt{xy}$ be a consumer's utility function, where x and y are the quantities of the two goods available to the consumer. The per unit price of the goods are \$5 (for X) and \$10 (for Y), respectively, and the consumer has \$500 available. Formulate and solve the optimization problem of the consumer.

Solution: The consumer's problem is

$$\max_{(x,y)\in\mathbb{R}^2_+} u(x, y) = \sqrt{xy} \qquad \text{s.t.} \quad 5x + 10y \leq 500, x \geq 0, y \geq 0.$$

Since $\sqrt{\cdot} : \mathbb{R} \to \mathbb{R}$ is a strictly increasing function, the solutions to the consumer's problem coincide with those of

$$\max_{(x,y)\in\mathbb{R}^2_+} xy \qquad \text{s.t.} \quad 5x + 10y \leq 500, x \geq 0, y \geq 0,$$

and we can solve the latter problem instead. Since the constraint functions are all affine, the KKT conditions are necessary.

We write down the KKT conditions

$$\begin{aligned}
&y - 5\lambda \leq 0, \quad x\,(y - 5\lambda) = 0, \quad x \geq 0\\
&x - 10\lambda \leq 0, \quad y(x - 10\lambda) = 0, \quad y \geq 0\\
&5x + 10y - 500 \leq 0, \quad \lambda(5x + 10y - 500) = 0, \quad \lambda \geq 0
\end{aligned}$$

If $\lambda = 0$ then by $y \leq 5\lambda$ and $x \leq 10\lambda$, $(x, y) = (0, 0)$. The point $(x^*, y^*, \lambda^*) = (0, 0, 0)$ satisfies the KKT conditions.

If $\lambda > 0$, by complementary slackness, $5x + 10y = 500$. Consequently, either $x > 0$ or $y > 0$, as otherwise $5x + 10y = 0$, a contradiction. If $x > 0$, complementary slackness implies $y = 5\lambda > 0$; similarly, if $y > 0$, complementary slackness implies $x = 10\lambda > 0$. Thus, it must be that $x > 0$ and $y > 0$. Again by complementary slackness, $y = 5\lambda$ and $x = 10\lambda$. Thus, $(x^*, y^*, \lambda^*) = (50, 25, 5)$ also satisfies the KKT conditions.

Since the constraint set (i.e., the budget set) is compact and the objective function is continuous, the extreme value theorem applies, and therefore a solution exists. Then since the KKT conditions are necessary; that is, every solution must (together with some $\lambda^* \geq 0$) satisfy the KKT conditions, it suffices to compare the values of the objective function at $(50, 25)$ and $(0, 0)$. Since $u(50, 25) = 1250 > 0 = u(0, 0)$, we can conclude that $(x^*, y^*) = (50, 25)$ is the unique solution to the consumer's problem.

Remark. For KKT conditions, even if there are only two variables and one constraint (excluding nonnegativity constraints), discussing the signs of x, y, and λ can be daunting. It is usually helpful to start with a subset of them. In this question, we start from discussing the sign of λ; in some other questions, discussing the signs of x and y first can also be helpful.

Problem 3

Find a solution to the maximization problem

$$\begin{aligned} \max_{(x_1,x_2)\in\mathbb{R}^2} \quad & -x_1^2 - x_1x_2 - x_2^2 \\ \text{s.t.} \quad & -x_1 + 2x_2 \geq 1 \\ & 2x_1 + x_2 \leq 2. \end{aligned}$$

(*Hint.* Use the trick used in Tutorial 8 Problem 1 to prove that the objective function is concave. If you have difficulties proving this fact, assume that it is concave and move on.)

Solution: Denoting the objective function by $f(x_1, x_2)$, note that it can be written as

$$f(x_1, x_2) = -\left(x_1 + \frac{1}{2}x_2\right)^2 - \frac{3}{4}x_2^2.$$

Denote $f_1(x_1, x_2) = -\left(x_1 + \frac{1}{2}x_2\right)^2$, and $f_2(x_1, x_2) = -\frac{3}{4}x_2^2$. Letting $g(x_1, x_2) = x_1 + \frac{1}{2}x_2$ and $h(z) = -z^2$, we know that g is affine and h is concave. By tutorial 8 problem 1 part (a), since $f_1 = h \circ g$, f_1 is concave. Since f_2 is also concave, and the sum of two concave functions is concave, the objective function must be concave.

Because the objective function is concave and the constraints are linear, we know that the Lagrange conditions are necessary and sufficient; that is, a point (x_1^*, x_2^*) is a solution to the maximization problem if and only if there exist Lagrange multipliers $\lambda_1^*, \lambda_2^* \geq 0$ such that $(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*)$ satisfies the Lagrange conditions.

To write down the Lagrange conditions, it is useful to write the first constraint as $x_1 -$

$2x_2 \leq -1$. The Lagrange conditions are

$$\begin{gathered} -2x_1 - x_2 - \lambda_1 - 2\lambda_2 = 0 \\ -x_1 - 2x_2 + 2\lambda_1 - \lambda_2 = 0 \\ \lambda_1 \geq 0, \quad x_1 - 2x_2 + 1 \leq 0, \quad \lambda_1 (x_1 - 2x_2 + 1) = 0 \\ \lambda_2 \geq 0, \quad 2x_1 + x_2 - 2 \leq 0, \quad \lambda_2 (2x_1 + x_2 - 2) = 0. \end{gathered}$$

As usual, we start by discussing the signs of λ_1 and λ_2.

- $\lambda_1 = 0, \lambda_2 = 0$.

  The first two equalities then reduce to

  $$\begin{aligned} -2x_1 - x_2 = 0, \\ -x_1 - 2x_2 = 0. \end{aligned}$$

  But this means that $x_1 = x_2 = 0$, which violates the first constraint. A contradiction.

- $\lambda_1 > 0, \lambda_2 = 0$.

  Since $\lambda_1 > 0$, by complementary slackness, $x_1 - 2x_2 = -1$. Thus, $x_1 = 2x_2 - 1$; plugging this and $\lambda_2 = 0$ into the first two equalities, we get

  $$\begin{aligned} -4x_2 + 2 - x_2 - \lambda_1 = 0 \\ -2x_2 + 1 - 2x_2 + 2\lambda_1 = 0 \end{aligned}$$

  which can be simplified to

  $$\begin{aligned} 5x_2 + \lambda_1 = 2; \\ 4x_2 - 2\lambda_1 = 1. \end{aligned}$$

  Solving this system of equations, we get $x_2 = 5/14$ and $\lambda_1 = 3/14$; then $x_1 = 2x_2 - 1 = -2/7$. Thus, $(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*) = (-2/7, 5/14, 3/14, 0)$ satisfies the Lagrange conditions.

- $\lambda_1 = 0, \lambda_2 > 0$.

  Similar to the previous case, since $\lambda_2 > 0$, by complementary slackness, $2x_1 + x_2 = 2$, and hence $x_2 = 2 - 2x_1$. Plugging this and $\lambda_1 = 0$ into the first two equalities, we get

  $$\begin{aligned} -2x_1 - 2 + 2x_1 - 2\lambda_2 = 0, \\ -x_1 - 4 + 4x_1 - \lambda_2 = 0. \end{aligned}$$

  But solving this system yields $\lambda_2 = -1 < 0$, a contradiction.

- $\lambda_1 > 0, \lambda_2 > 0$. Complementary slackness implies that

  $$\begin{aligned} x_1 - 2x_2 = -1, \\ 2x_1 + x_2 = 2. \end{aligned}$$

  Solving this system, we get $x_1 = 3/5$ and $x_2 = 4/5$. However, since $\lambda_1, \lambda_2 > 0$, the condition $-2x_1 - x_2 - \lambda_1 - 2\lambda_2 = 0$ must be violated.

Thus, $(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*) = (-2/7, 5/14, 3/14, 0)$ is the unique point satisfying the Lagrange conditions. Then since the Lagrange conditions are necessary and sufficient, $(x_1^*, x_2^*) = (-2/7, 5/14)$ is the unique solution of the maximization problem.